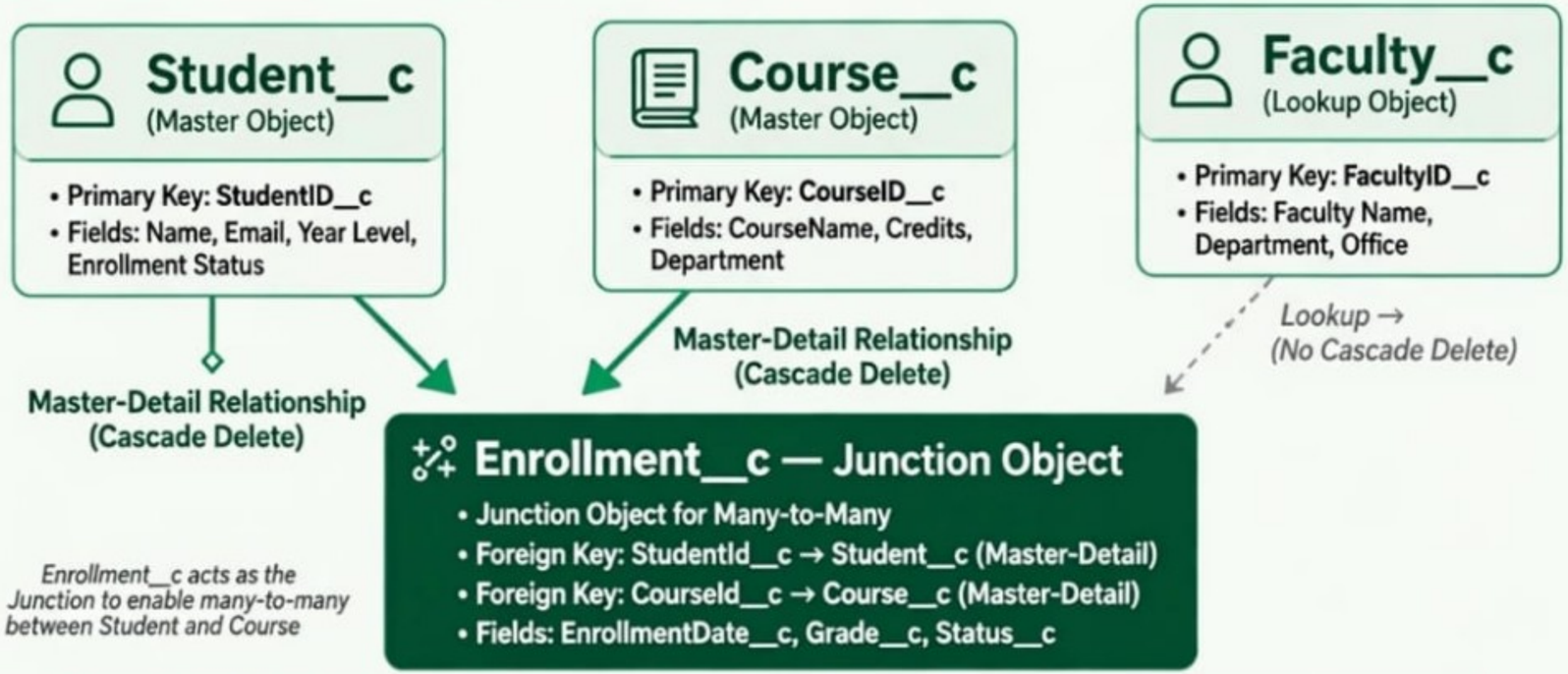
RELATIONSHIPS DEEP DIVE — ER DIAGRAM & DATA MODEL for 2026-2027

ER DIAGRAM
RELATIONSHIPS OVERVIEW



RELATIONSHIP DETAILS

Master-Detail: Student → Enrollment & Course → Enrollment

- Child object is Enrollment__c, tied to two Masters:
- Required relationship — Enrollment cannot exist without a Student AND a Course
- Roleup summaries available: e.g., Count of Enrollments on Student

Lookup: Faculty → Course

- Optional relationship between Faculty and Course
- A Course can be assigned to one Faculty as Instructor
- Deleting a Faculty does NOT delete the Course record — Course remains
- Allows flexible reassignment of Faculty later

MANY-TO-MANY VIA JUNCTION OBJECT

- Student ↔ Course is many-to-many relationship
- Enrollment__c is the Junction object connecting them
- One Student can enroll in many Courses, one Course can have many Students

WHY MASTER-DETAIL WAS CHOSEN FOR ENROLLMENT?

Data Integrity

Ensures every Enrollment record must reference valid Student and Course. Prevents orphaned records & maintains consistency.

Cascade Delete for Cleanup

If a Student or Course is deleted, all related Enrollment records are automatically deleted. No manual cleanup required, avoids orphaned data.

Security & Roll-up Summaries

Child inherits sharing from parent for security. Enables Roll-up Summary fields on Student/Course to track total enrollments, avg grade, etc.

FOREIGN KEYS & CASCADDE DELETE EXPLAINED:

Enrollment__c uses two Foreign Keys: StudentId__c (Master-Detail FK to Student__c) and CourseId__c (Master-FK to Course__c). These FKs form the composite key. Cascade Delete rule means deleting the parent Student or Course automatically deletes the related Enrollment records, keeping database clean.

STUDENT ENROLLMENT SYSTEM

Phase 5 • Project Development Phase • Academic Year 2026–2027

BUILDING STAGE — Custom Objects & Fields Overview

This is the actual building stage: designing the custom data model for the Student Enrollment System. Four custom Salesforce objects are being created to support student enrollment, course management, and faculty assignments.

Student__c

Purpose: Stores student profile and contact information

Field	Data Type	Description
Email	Formula (Formula, Text)	Auto-populated email address using formula: FirstName & '.' & LastName & @university.edu
Phone	Text (Validation Rule)	Validation: 10-digit phone number format (123) 456-7890
Status	Picklist	Values: Pending • Approved • Waitlist

Course__c

Purpose: Manages course offerings and details

Field	Data Type	Description
Course Capacity	Number (60)	Maximum number of students per course. Set to 60 as default limit.
Course Name	Text	Name of the course (e.g., MATH 101, CS 202)
Credit Hours	Number	Academic credit hours for the course

Enrollment__c (Junction)

Purpose: Junction object linking Students to Courses (Many-to-Many)

Field	Data Type	Description
Status	Picklist	Tracks enrollment status: Pending • Approved • Waitlist
Grade	Text	Final grade recorded after course completion (e.g., A, B, C, Pass/Fail)
Student__c	Lookup (Course)	Lookup relationship to Gtudent__c object

Faculty__c

Purpose: Stores faculty/instructor information

Field	Data Type	Description
Email	Formula (Formula, Text)	Auto-generated faculty email: FirstName & '." & LastName & @faculty.university.edu
Phone	Text (Validation Rule)	Validation: 10-digit phone number with area code required
Department	Text	Faculty relationsent assignment (e.g., Computer Science, Mathematics)

Implementation Notes — Building Stage

- All email fields use Salesforce Formula data type to ensure consistent university email format
- Phone fields include validation rule to enforce standard 10-digit phone format
- Enrollment__c acts as junction to handle many-to-many relationship between Student and Course

Key Focus: Data Model Design

- Relationships established using Lookup fields for data integrity
- Picklist ensures controlled status values for enrollment
- Capacity set to 60 students per course for scalability

STUDENT ENROLLMENT SYSTEM — PHASE 5: PROJECT DEVELOPMENT PHASE

Academic Year 2026–2027

AUTOMATION

RECORD-TRIGGERED FLOW: AUTO EMAIL AFTER ENROLLMENT

1. **Record Created** — Enrollment__c Record Created
2. **Evaluate Criteria** — Is Status = "Enrolled"?
3. **Get Related Course** — Retrieve Course Record & Capacity
4. **Send Email Action** — Send Confirmation Email to Student
5. **Email Sent** — Confirmation Email Delivered

Flow Type: Record-Triggered Flow | Object: Enrollment__c | Trigger: When record is created | Condition: Status = Enrolled

APEX TRIGGER

BLOCK ENROLLMENT IF COURSE CAPACITY FULL

EnrollmentTrigger.apxc —

```
trigger EnrollmentTrigger on Enrollment__c (before insert) {
 // Check if course is at 60 seat capacity before insert
-  for (Enrollment__c newEnroll : Trigger.new) {
-      Integer currentCount = [
       SELECT COUNT() FROM Enrollment__c
       WHERE Course__c = :newEnroll.Course__c
-      ];
-     if (currentCount >= 60) {
        newEnroll.addError(
        "Capacity Full: This course has reached 60 seats. Enrollment closed."
-      );
```

Logic: Query current enrollments for the course → If count ≥ 60 then addError() to block the record

Purpose:

- Prevents over-enrollment beyond the 60-seat limit
- Enforces data integrity at database level for rellability

UI — LIGHTNING APP & STUDENT PORTAL (LIGHTNING WEB COMPONENTS LWC)

Enrollment Form

LWC Component

Student ID: STU-10045

Course: CS101 — Intro to Programming — 58/60 seats

Submit Enrollment

Course Catalog

LWC Component

- **CS101** Intro to Programming — 58/60 Enrolled — Open
- **MATH202** Calculus II 45/60 Enrolled — Open
- **PHYS101** Physics I 60/60 Enrolled — Full

Student Dashboard & Faculty Grading

LWC

Your Enrollments: 2 Courses • GPA: 3.6 — 60%

Faculty Grading View:

- CS101 — 22 Submissions Pending
- MATH202 — 15 Grades Posted

KEY BENEFITS — PHASE 5 DEVELOPMENT OUTCOMES

- **Reduced Manual Work** — Automated emails remove manual follow-ups, saving 10+ hours per week
- **Real-time Capacity Enforcement** — Apex trigger prevents overbooking and enforces the 60-seat capacity limit
- **Improved Student Experience** — Instant email confirmation and self-service portal for students
- **Transparency & Reporting** — Faculty & Admins gain real-time dashboard insights and reporting